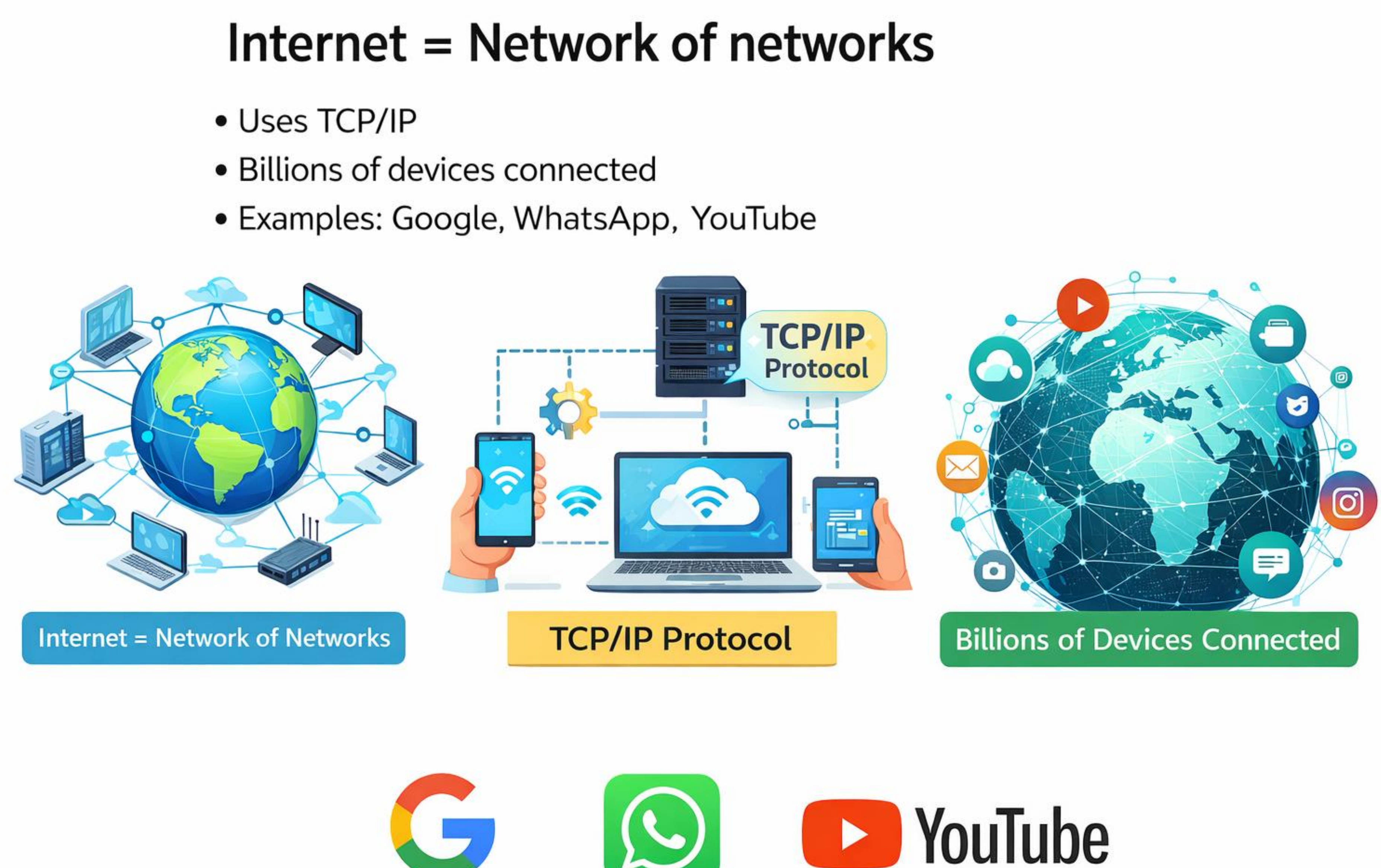
Internet Fundamentals

Internet & Web Concepts

Internet Fundamentals



Real-Time Example: Internet

Sending WhatsApp Message:

1. Message typed on phone
2. Converted to data packets
3. Sent through Internet
4. Delivered to receiver

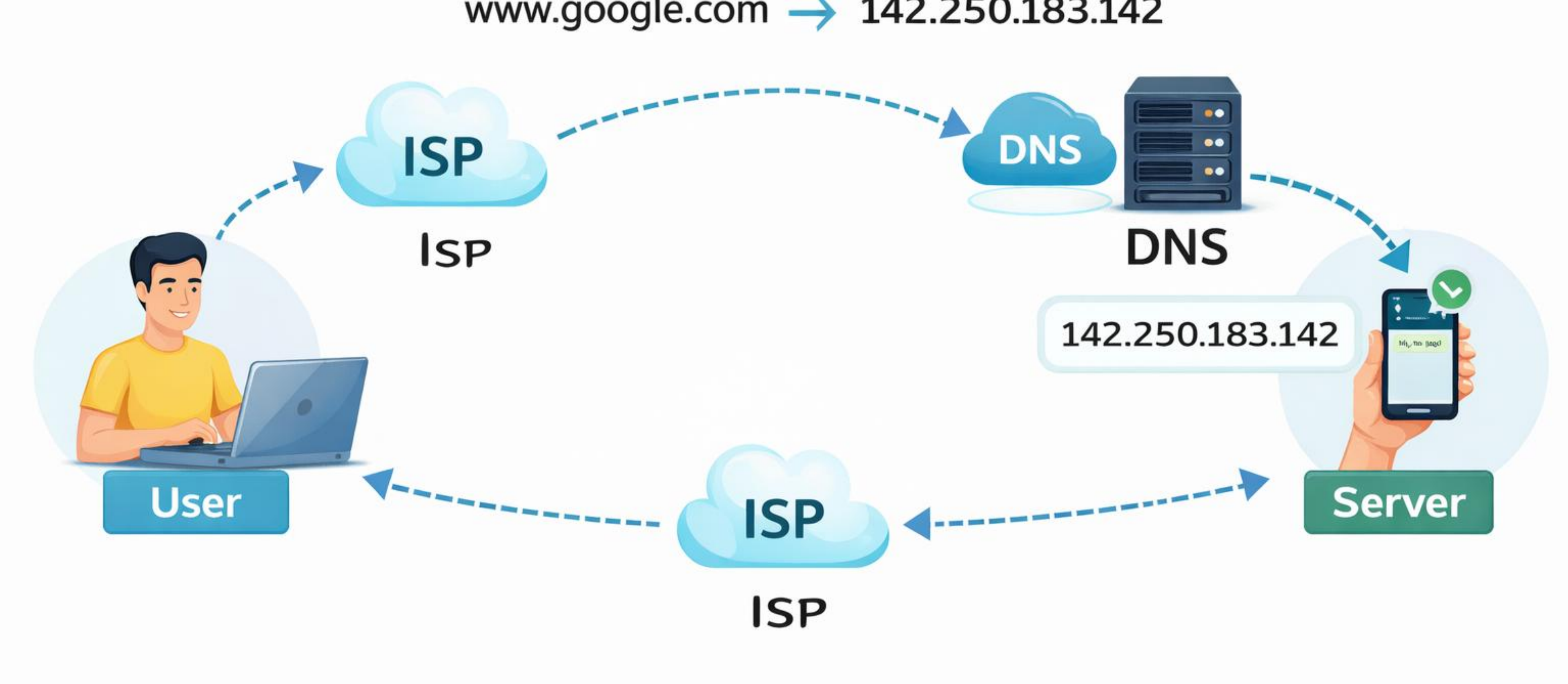
Client–Server Architecture

- **Client:**
- Browser, Mobile App

- **Server:**
- Stores data
- Responds to requests

Client–Server Analogy

- ATM Transaction:
- Client: ATM Machine
- Server: Bank Server
- Request: Withdraw money
- Response: Cash / Error



How Internet Works (Flow)

- User → ISP → DNS → Server
- Server → ISP → User
- DNS converts domain name to IP address

🌐 What is ISP?

ISP = Internet Service Provider

An **ISP is a company that gives you access to the Internet.**

Without an ISP, your mobile or computer **cannot connect to the Internet.**

What does an ISP do?

- Connects your device to the Internet
- Assigns an **IP address** to your device
- Sends and receives data from websites and servers
- Acts as a **bridge between you and the Internet**

- **Analogy:**

Think of ISP as a **road provider.** If there is no road, your vehicle cannot travel. Similarly, **without ISP, data cannot travel**.

- **Examples of ISPs:**

Mobile ISPs: Jio, Airtel, Vi (Vodafone Idea)

Broadband ISPs: ACT Fibernet, BSNL, Hathway

What is DNS?

DNS = Domain Name System

DNS **converts website names into IP addresses.**

Computers understand **numbers (IP addresses)**, but humans remember **names (google.com)**.

Why DNS is Needed?

- Humans type: www.google.com
- Computer needs: 142.250.183.142
- DNS does this **conversion automatically**

Analogy:

Think of DNS as a **phone contact list** . You save: **“Mom”.** Phone dials: **actual phone number**

Similarly:

You type: **google.com**

DNS finds: **IP address of Google**

Examples of DNS Servers:

Google DNS → 8.8.8.8

Cloudflare DNS → 1.1.1.1

ISP DNS → Provided by Jio, Airtel, etc.

🔄 How ISP and DNS Work Together (Flow)

- User types www.youtube.com
- Request goes to **ISP**
- ISP asks **DNS** → "What is the IP address?"
- DNS replies with IP address
- ISP connects user to **YouTube server**
- Website loads on browser

Summary:

ISP→ Internet Provider

DNS → Internet Phone Book

IP Address → Internet House Number

Web Browser

*Software to access WWW

*Sends requests to websites

*Receives web pages

*Interprets HTML

*Displays webpages in a readable form

The browser **does NOT store websites
It only **requests and shows** them

Web Browser

- Web Browser is a software to see web pages.

World Wide Web (WWW)

- WWW runs on Internet
- Uses HTTP/HTTPS
- Web pages written in HTML
- Accessed using browsers

**World Wide Web is not the Internet itself.
It is a service that runs *on top of* the Internet, just like apps run on your phone.

Internet vs WWW

- Internet = Network infrastructure
- WWW = Collection of websites

- **Analogy :**

Internet is like **roads**.
WWW is like **vehicles using those roads**.
Roads can exist without vehicles, but vehicles need roads.”

HTTP / HTTPS (Communication Rules)

- HTTP = HyperText Transfer Protocol

➢ Stateless

➢ Request–Response Model

➢ Methods: GET, POST

- HTTPS = Secure HTTP

➢ Secure HTTP

➢ Uses SSL/TLS

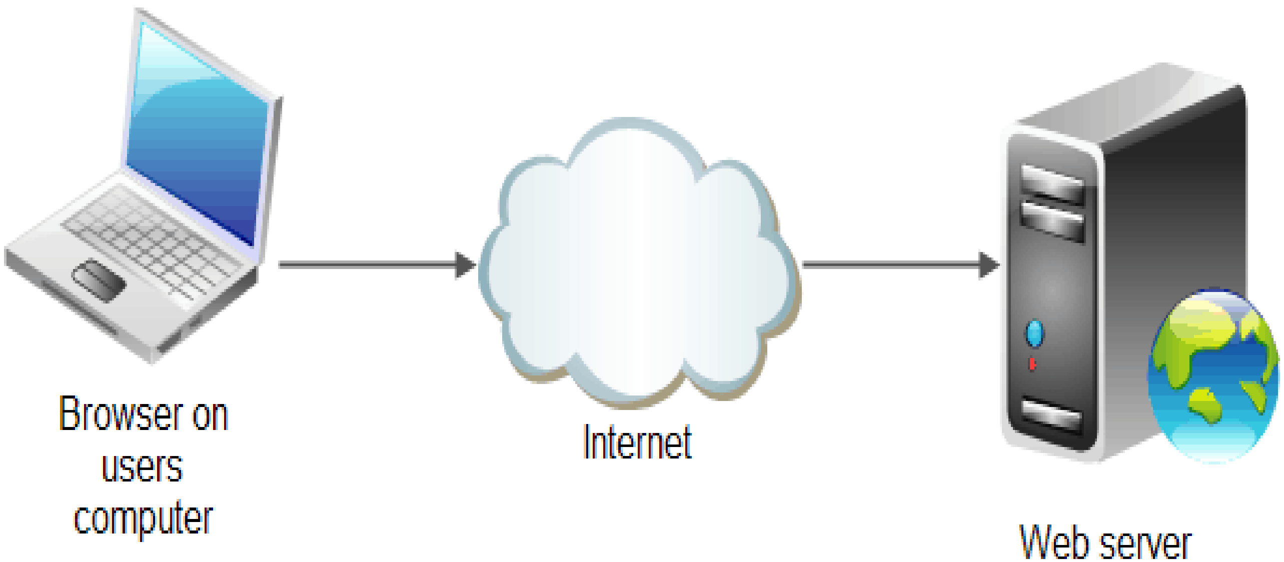
➢ Encrypts data

- Used between browser and server

- “HTTP is like the **language** used for conversation. HTTPS is the same language, but spoken inside a **locked room**.”
- **Real example:**
- Online banking → HTTPS
- Normal blog → HTTP/HTTPS

Web Server

- Web Server is a computer where web pages are kept. It provides requested resources as service so called Server.



HTTP Request Example

- GET /login HTTP/1.1
- Host: college.edu
- User-Agent: Chrome

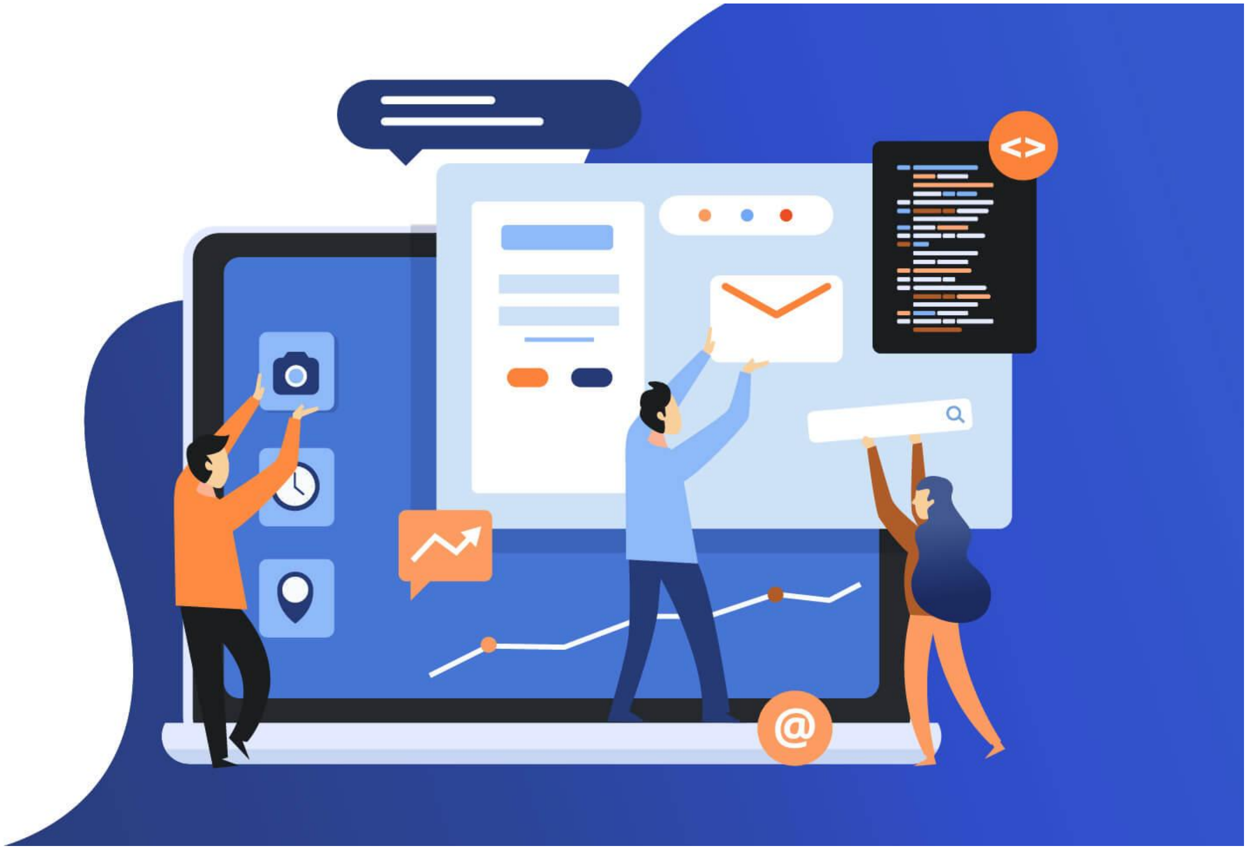
HTTP Response Example

- HTTP/1.1 200 OK
- Content-Type: text/html

- <h1>Welcome Student</h1>

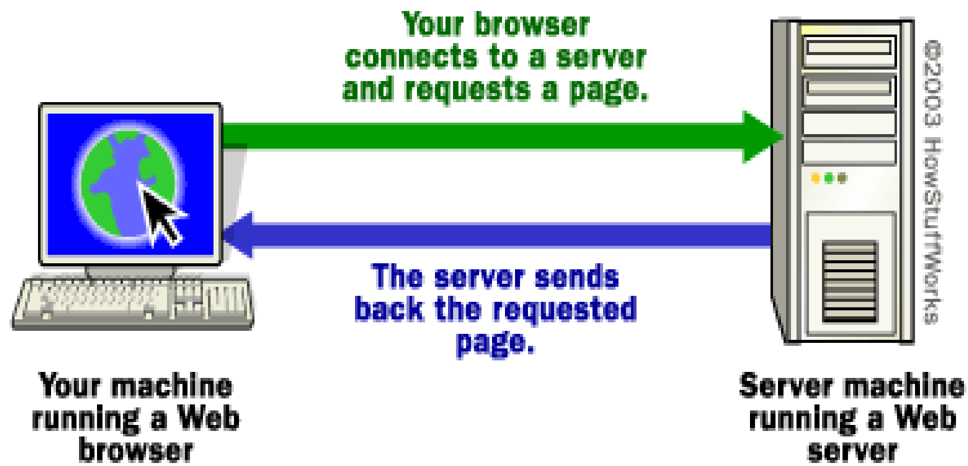
Web Site

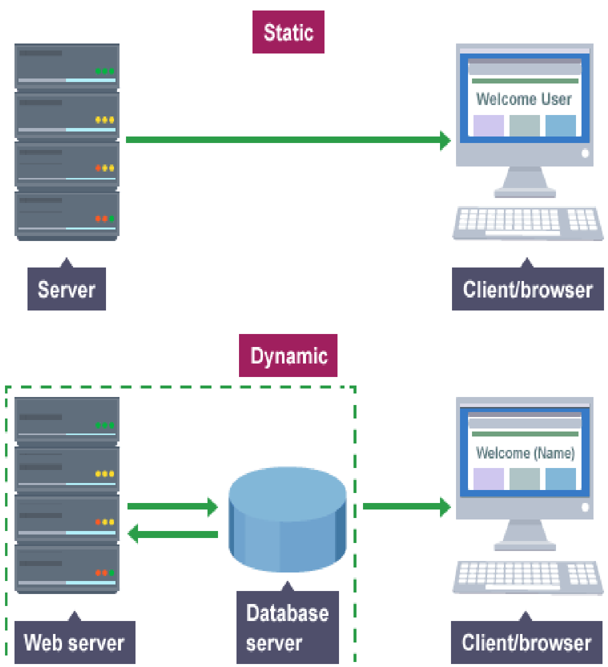
- Web site is a collection of related web pages from the similar matter or domain.

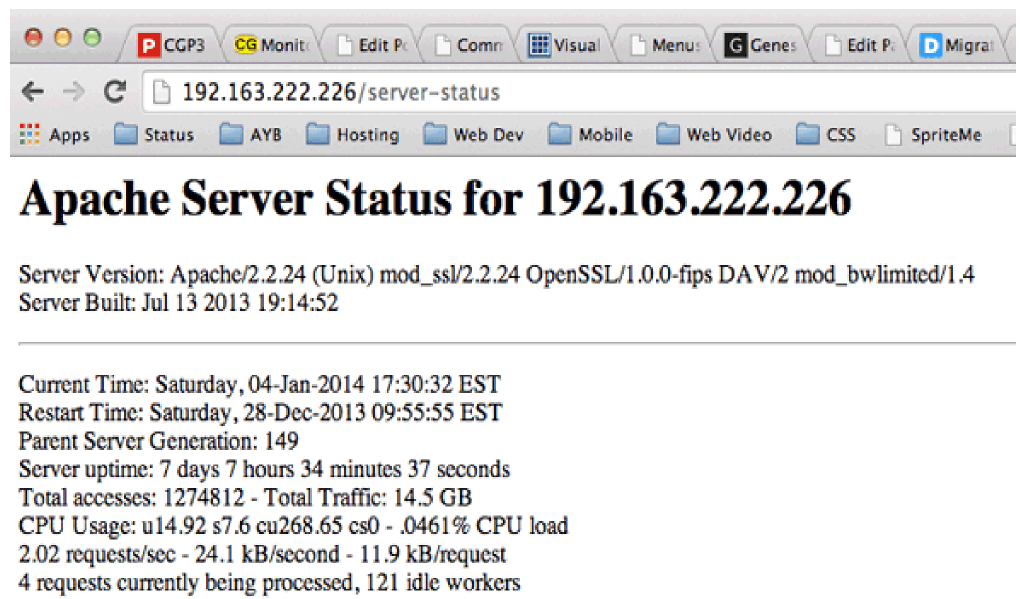


Types of Web Pages

- Static
- Dynamic
- Server page







Apache Server Status for 192.163.222.226

Server Version: Apache/2.2.24 (Unix) mod_ssl/2.2.24 OpenSSL/1.0.0-fips DAV/2 mod_bwlimited/1.4
Server Built: Jul 13 2013 19:14:52

Current Time: Saturday, 04-Jan-2014 17:30:32 EST
Restart Time: Saturday, 28-Dec-2013 09:55:55 EST
Parent Server Generation: 149
Server uptime: 7 days 7 hours 34 minutes 37 seconds
Total accesses: 1274812 - Total Traffic: 14.5 GB
CPU Usage: u14.92 s7.6 cu268.65 cs0 - .0461% CPU load
2.02 requests/sec - 24.1 kB/second - 11.9 kB/request
4 requests currently being processed, 121 idle workers

HTML – Web Page Language

HTML defines structure of web pages

Used to create text, images, links

"HTML is like a **house blueprint**.
It does not decorate, but it tells where everything should be."

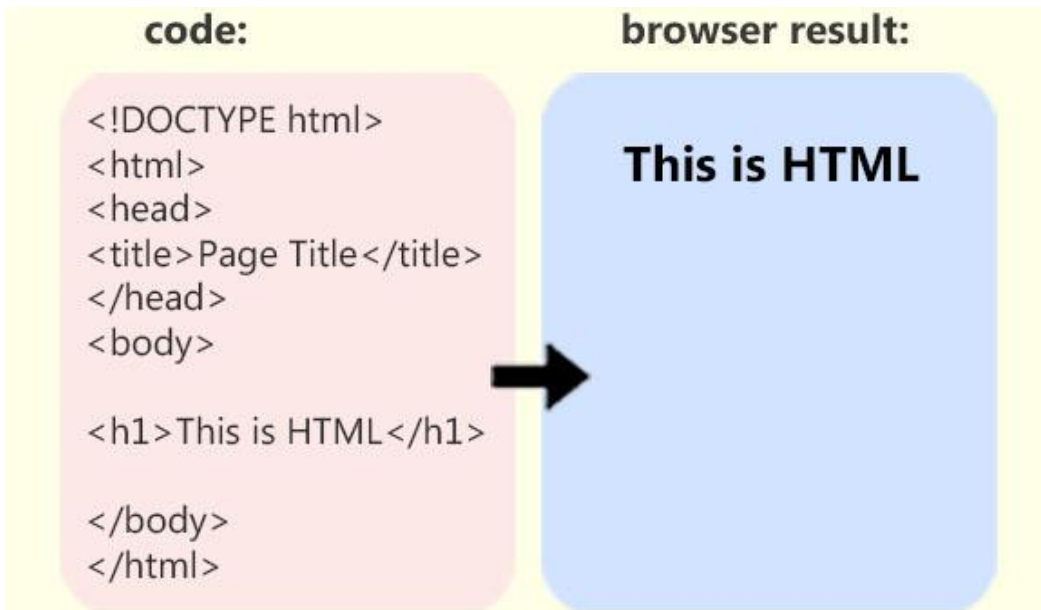
HTML → Structure

CSS → Design

JavaScript → Actions

What is HTML?

- HTML stands for Hyper Text Mark-up Language, a standard markup language for creating web pages.
- Defines structure and content of a webpage
- Uses tags to represent elements



How HTML Works

- HTML File → Browser → Web Page

- Browser reads HTML code
- Interprets HTML tags
- Renders content visually

- <h1> → Heading
- <p> → Paragraph
- <a> → Link

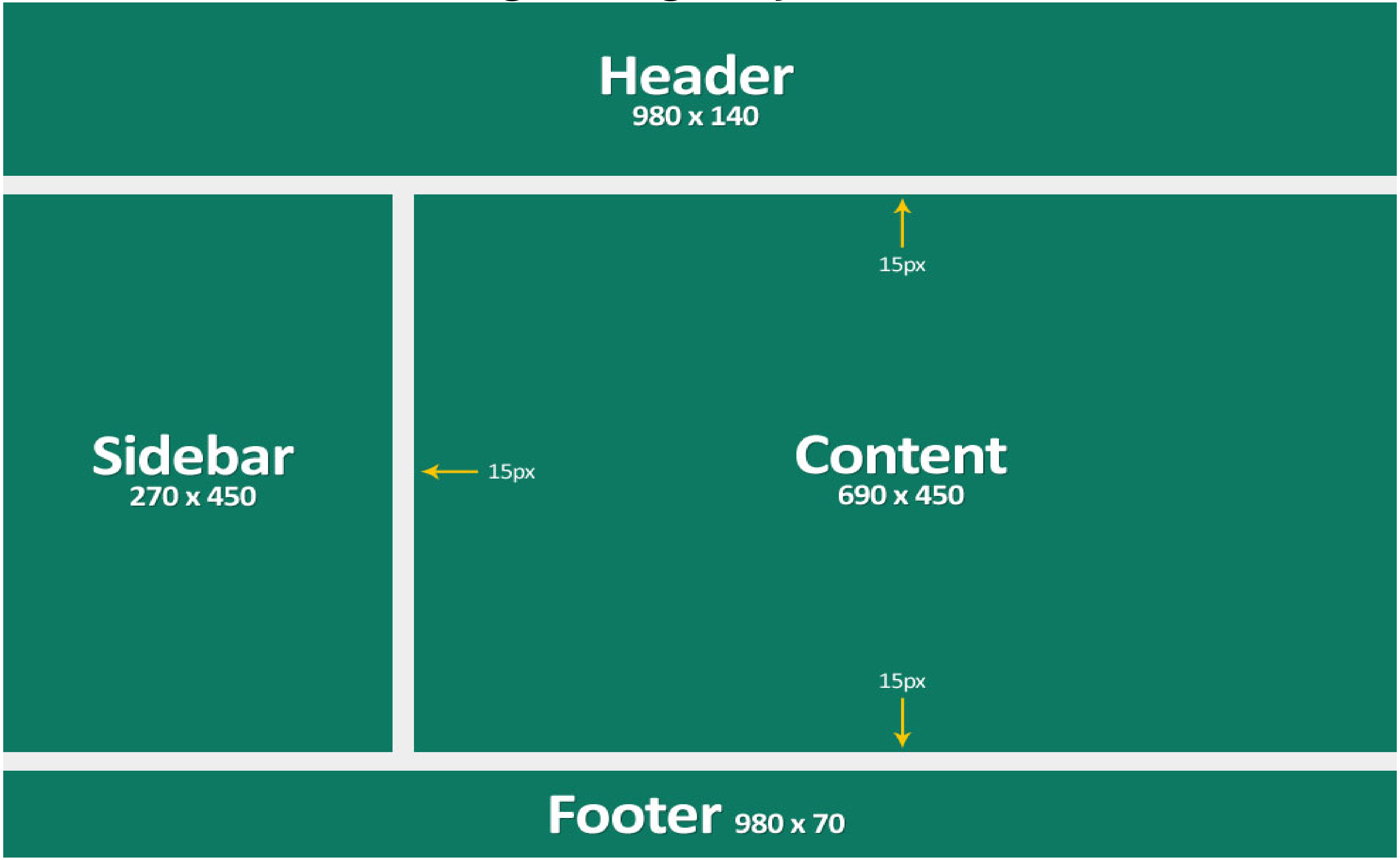
Role of HTML in Web Development

- HTML works with other technologies:

- HTML → Structure
- CSS → Styling & Layout
- JavaScript → Interactivity

- HTML is the foundation of every webpage.

Basic HTML Web Page Design Layout



Analogy

Roads & Physical cables – Internet :(Infrastructure)

Cable network or DTH network – WWW (made for websites, uses HTTP/HTTPS rules). :(service)

Television = Browser (Displays content, Cannot create content, Just shows what is broadcast) : (Tool)

Channel = Website (Google, You Tube): (content)

Program = Web page (Home Page, About Page, Login Page).

Broadcast Language = HTTP/HTTPS (Rules for sending web data, HTTPS = Secure Broadcast

STEPS

🔹 STEP 1: Browser Sends Request

What happens?

1. User types **amazon.in** in the address bar

2. Browser understands:
"I need Amazon's home page"

3. Browser sends an **HTTP/HTTPS request**:
•GET / HTTP/1.1 Host: amazon.in

•This request goes through:
•ISP
•Internet
•Amazon server

Browser acts as a **client

It always **initiates communication

Uses **HTTP or HTTPS

STEPS

- 🔷 **STEP 2: Server Sends Webpage**
- **What happens at Amazon Server?**
- Amazon server receives request
- Server processes it
- Server sends back:
  - HTML file
  - CSS files
  - Images
  - JavaScript

- All these are sent as **data packets**
- Server is always **listening**
- Server only responds when requested
- Server sends **raw data**, not visuals

STEPS

- 🔹 **STEP 3: Browser Renders Content**
- **This is the MOST IMPORTANT browser function**
- **What does “render” mean?**
- Convert **code into visual page**

RENDERING

- A **rendering engine** is the **part of the browser that converts web code into a visible webpage**.

- **Rendering engine = Code → Visual page**
- Without a rendering engine, the browser would only receive **text files**, not webpages.

Rendering steps

- **Step 1: Read HTML (Parsing)**

Browser receives HTML from server

Rendering engine **reads the HTML line by line**

Converts HTML into a tree structure called **DOM**

DOM = Document Object Model

Think of DOM as a **family tree** of HTML elements

- **Step 2: Read CSS (Style Calculation)**
- Rendering engine reads CSS files
- Figures out:
  - Colors
  - Fonts
  - Sizes
  - Positions
- Creates a **CSSOM** (CSS Object Model)

Rendering steps

Step 3: Create Render Tree

- DOM + CSSOM are combined
- Invisible elements are removed
- Final **Render Tree** is created
- Render Tree contains **only what should be displayed**

WHY RENDERING ENGINE IS IMPORTANT

- ✓ Converts code to visuals
  ✓ Enables fast page loading
  ✓ Supports animations & interactivity
  ✓ Makes browsing possible

- “Rendering engine is the heart of the browser.”

Internally what does browser do

File Type	Browser Action
HTML	Creates structure
CSS	Adds colors & design
JavaScript	Adds actions
Images	Displays pictures

Final Output:

- Amazon homepage
- Product images
- Buttons
- Search bar

- Summary:

User → Browser → Internet → Server
Server → Internet → Browser → Screen

Router

- A **router** is a **network device** that **directs data to the correct destination** on the Internet.
- It decides **where data should go next**.
- Just like:
- Post office reads address and sends letter
- Router reads IP address and sends data

<u>Router - Post office sorter:</u>
Many roads exist
Router checks the **address**
Sends data on the **best path**

What Does a Router Do?

- Connects different networks
- Reads **IP addresses**
- Forwards data packets
- Chooses best path (routing)

- **Example (Home Wi-Fi)**
- At your home:
- Mobile , Laptop → Wi-Fi Router
- Router connects to ISP
- Router sends data to Internet and back
- Your router is the **gateway to the Internet**.

Where Are Routers Used?

- Home Wi-Fi
- College networks
- ISP networks
- Internet backbone
- Internet is made of **millions of routers**.

What Are **Data Packets**?

- A **data packet** is a **small piece of data** sent over a network.
- Big data is broken into **small packets** before sending.

- **Why Packets Are Needed?**

Faster transmission

Less congestion

Reliable delivery

Easy error handling

Example: WhatsApp Message

- Message: **"Hi"**
- Message is converted to binary
- Split into packets
- Each packet travels separately
- Packets reassembled at receiver
- Routers do NOT send messages.
  They send packets.
- Each router: Reads packet address, Forwards packet
- Packets carry data, routers guide packets.”

MCQs

- 1. Which protocol is secure?
- a) HTTP b) FTP c) HTTPS d) SMTP

- 2. Client example?
- a) Browser b) Server c) Router d) DNS

QUIZ

- **Q. Is browser same as Internet?**
  ❌ No
  ✔️ Browser is software; Internet is network.
- **Q. Can we access web without browser?**
  ✔️ Yes (curl, Postman, apps)

Final Summary

- Internet connects devices
- WWW uses HTTP/HTTPS
- Client–Server is core model
- Code shows real working